Automated Sentiment Analysis for Library Assessment

Using Machine Learning to Evaluate Student Feedback on Library Services

Sinbad Adjuik
Likitha Gindi

Problem and Objective

Problem:

- Academic libraries collect large amounts of student feedback through surveys, comments, and reviews. Manually analyzing this feedback is time-consuming and difficult to scale.

Objective:

- Develop a machine learning model that automatically classifies the sentiment of student feedback (positive, neutral, negative) to help libraries identify trends and improve services.

Key Question:

- Can machine learning accurately predict student sentiment toward library services?

Methodology and Expected Outcomes

Methodology:

- Text preprocessing (tokenization, stop-word removal, lemmatization).
- Feature extraction using TF-IDF.
- Integration of a pre-trained AI sentiment analysis model (e.g., BERT or DistilBERT) to enhance classification accuracy and reduce manual feature engineering.
- Model evaluation using Precision, Recall, F1 Score, and Confusion Matrix.

Data Collection & Sources

- Student surveys collected through online platforms
- Open-ended comments from library service forms
- Feedback from campus learning management systems
- Social media mentions related to library experiences

System Prototype / UI Overview

- Web-based dashboard for visualizing sentiment trends
  - Integrated chatbot for conversational feedback analysis
  - Admin panel for uploading new datasets
  - Real-time sentiment classification results displayed

Expected Outcomes:

Developed a web-based system with an AI-powered chatbot for conversational insights.

Identify overall sentiment trends in student feedback

Detect common positive and negative themes

Provide insights that support data-driven library service improvements